

Institute of Computer Engineering Technology

Coursework

Module	Programming Fundamentals
Name	Student Management System

Case Study

The Institute of Computer Engineering Technology (iCET) is a software engineering training institute in Sri Lanka, with over 1,000 students actively engaged in learning. Therefore, the iCET management has decided to upgrade its student management system to better handle student registration and monitor academic progress. The following use case outlines the system's scope and functionalities.

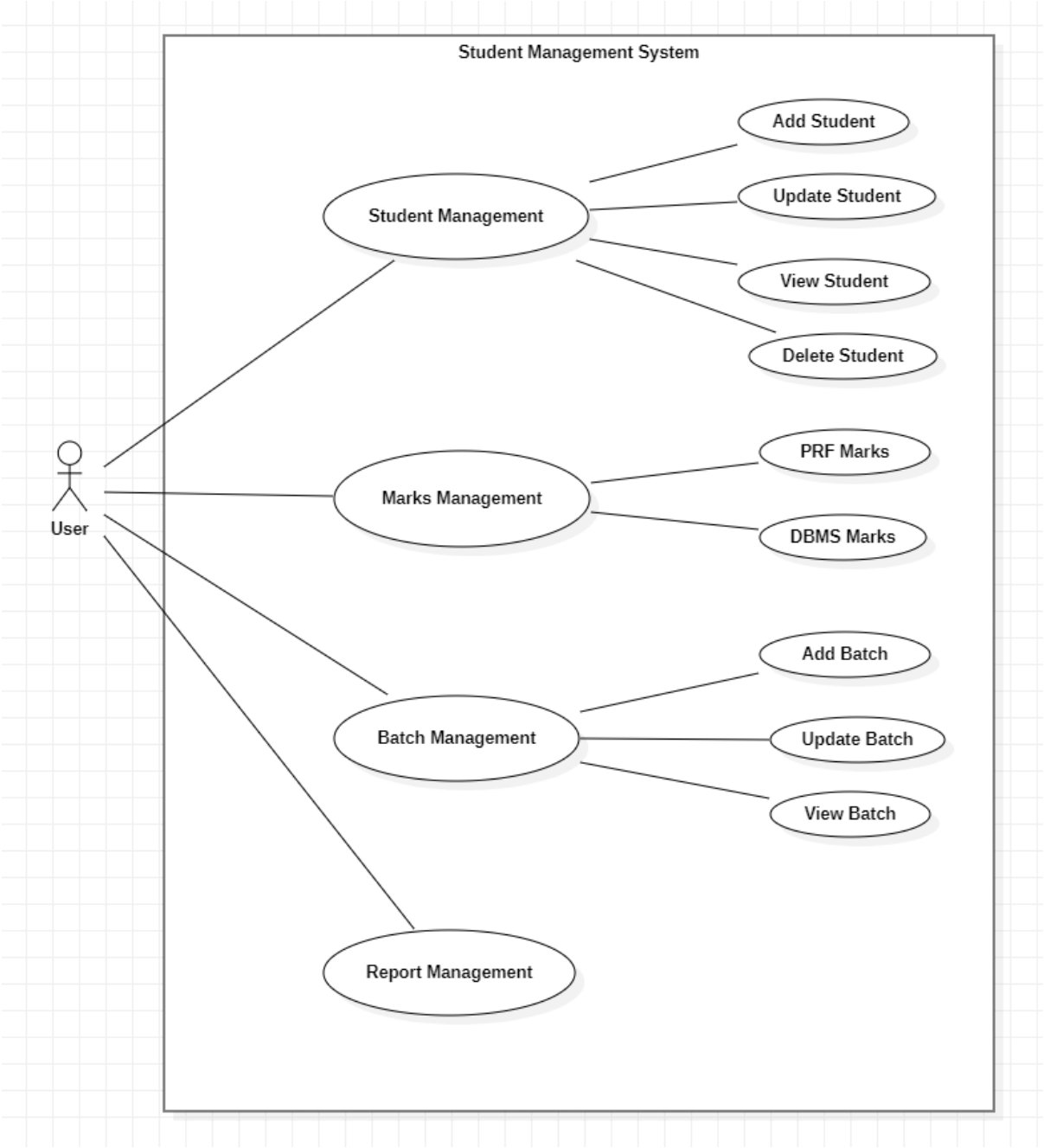


Figure 1 - iCET Student Management System UseCase Diagram

Requirements

You are required to create a Java application for a Student Management System. In this application, you need to implement the following use cases.

When the application is executed, it should display a Command Line Interface (CLI) similar to the example below, allowing the user to enter the number corresponding to the option they wish to execute. This will serve as the Home Page of the application you are developing.

```
-------------------------------------------------------------------------
|                      iCET Student Management System                   |
-------------------------------------------------------------------------

[1] Student Management

[2] Batch Management

[3] Grade Management

[4] Report Generator

[5] Exit

Enter an option to continue >
```

Figure 2 - Home Page of the iCET Student Management System

1. Student Management

There are four options in Student Management, and the user can select the corresponding option they wish to execute. Below is the Student Management interface you should develop.

```
-------------------------------------------------------------------
|                      Student Management                         |
-------------------------------------------------------------------

[1] Add Student

[2] Update Student

[3] View Student Profile

[4] Delete Student Profile

[5] Exit

Enter an option to continue >
```

Figure 3- Student Management Home Page

1.1 Add Student

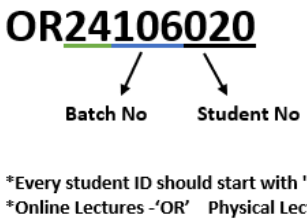
When adding a new student, the user must first select the batch to which the student will be added.

Batch Details:

- Batch Name – Each batch has a unique number starting from 100.
- Batch Status – There are two possible statuses for a batch:
    - ENROLLMENTOPEN = 1
    - ENROLLMENTCLOSED = 0

If the selected batch status is ENROLLMENTCLOSED, the system should prevent the user from adding students to that batch and inform the user accordingly. If the batch status is ENROLLMENTOPEN, the user can proceed to add a new student to the system. The system stores five key details for each student:

1. ***Student ID*** – This ID is generated automatically by the system and consists of 10 characters. If the student is registered for physical lectures, the ID should start with 'PR'. If the student is registered for online lectures, the ID should start with 'OR'.
   The remaining characters must follow a specific format and be sequential, based on the last Student ID in the selected batch. Student IDs must be unique and cannot be generated randomly.



2. ***Student NIC*** – The user must enter the student's NIC. The system uses the NIC to check whether the student already exists in the system. If the NIC is found, the system should display the existing student profile. If the NIC is not found, the system will prompt the user to enter the rest of the student details before adding the student.

3. ***Student Name*** – Student names can be duplicated.

4. ***PRF Marks*** – When adding a new student, the default marks should be set to **-2**, which indicates that the exam has not been conducted.

5. ***DBMS Marks*** – When adding a new student, the default marks should be set to **-2**, which indicates that the exam has not been conducted.

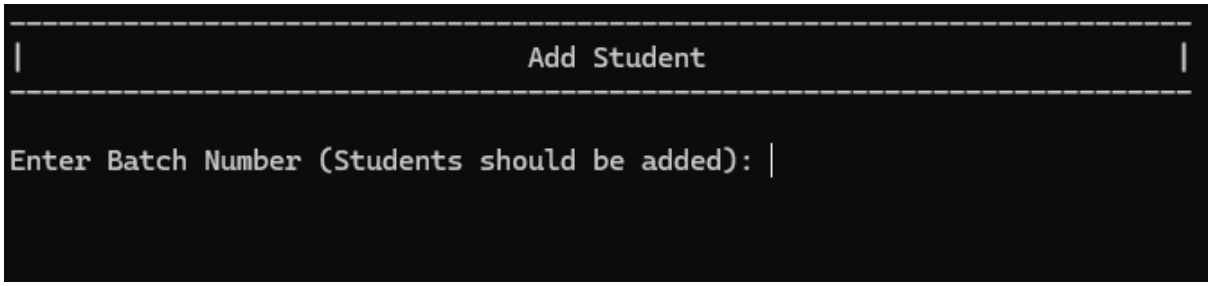


Figure 4 - Add New Student

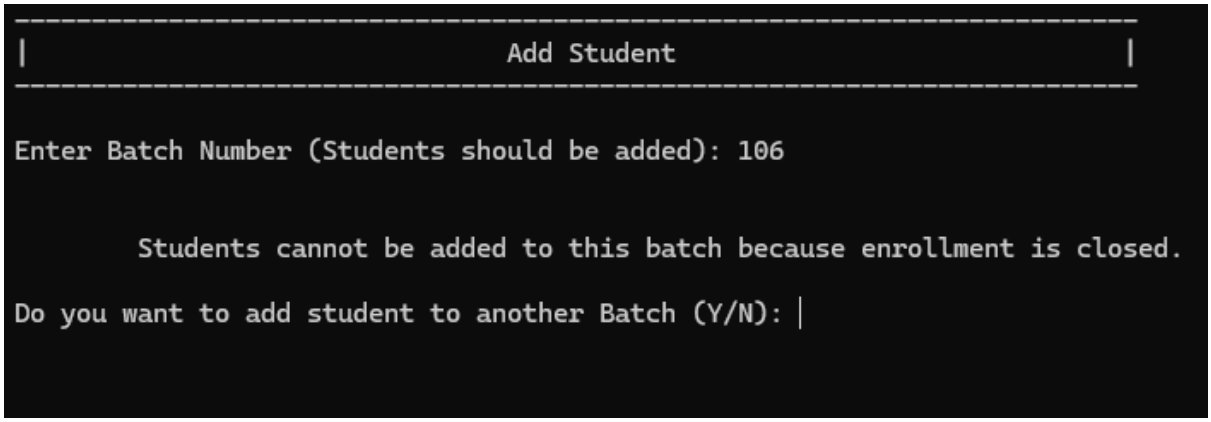


Figure 5 - Warning(Enrollment Closed)

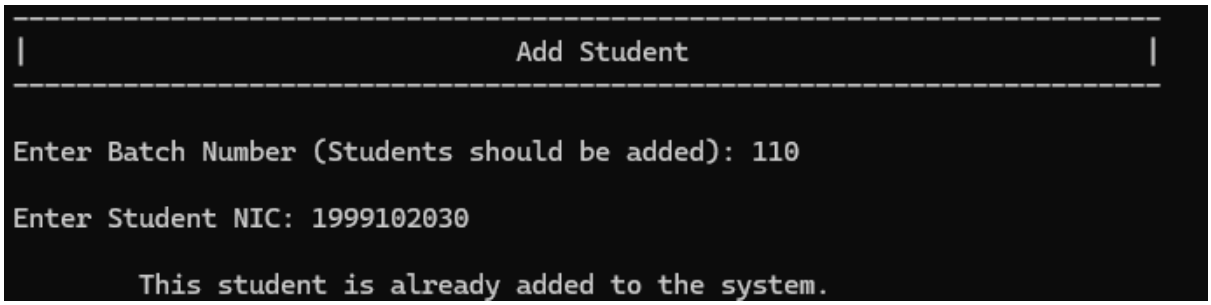


Figure 6 - Warning(Already added student)

After the student details are entered successfully, the system should add the student to the database and display the Student ID (student registration number). Once the student is added, the system should prompt the user with the message: “Do you want to add another student?” If the user enters “Y”, they can proceed to add another student. If the user enters “N”, they will be redirected to the home page.

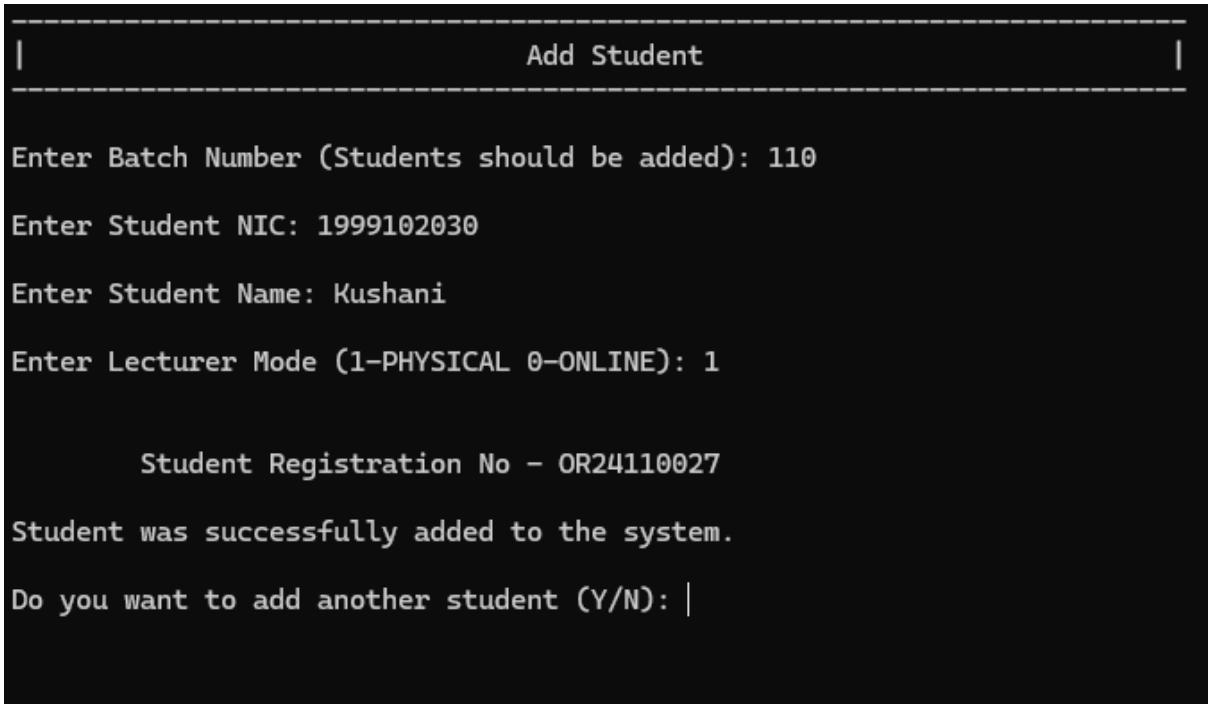


Figure 7 - Add New Student Successfully

1.2 Update Student

With this option, the user can update previously added student details (only the Name and NIC). First, the user needs to find the student by entering the Student ID. After the Student ID is entered, the system should check whether it exists in the system. If the Student ID has not been added yet, a message should be displayed as “This student not exist in the system”.

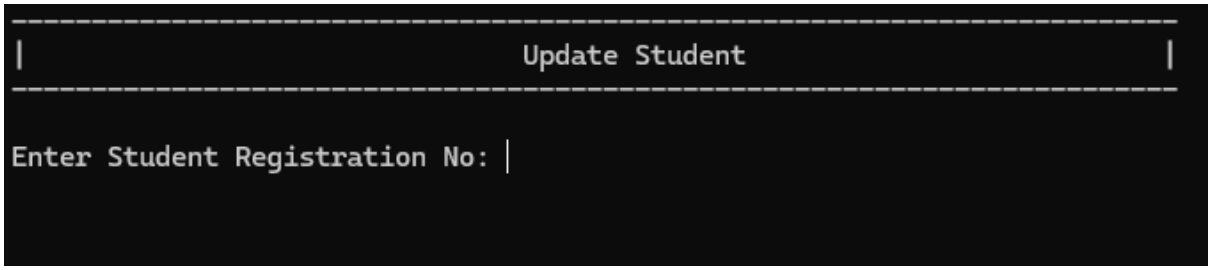


Figure 6 - Update Student

Once the user has entered a valid Student ID, the system will display the current details, which include the Student ID, Student Name, and Student NIC. After displaying the student details, the system should ask the user whether they want to update the Student Name or the Student NIC.

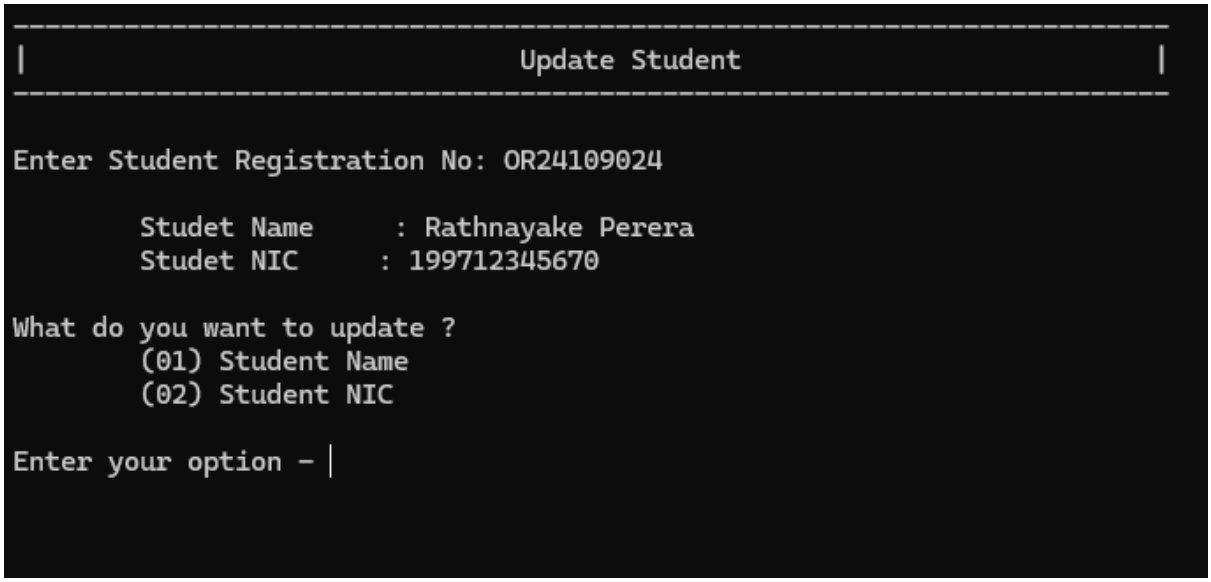


Figure 7 - Update Name or NIC

1.2.1 Student Name Update

If the user selects Student Name (1), a window related to the Student Name should be prompted, allowing the user to update the Student Name. Once the update has been completed successfully, the system should display a success message and prompt the user to choose whether to continue updating or return to the main menu.

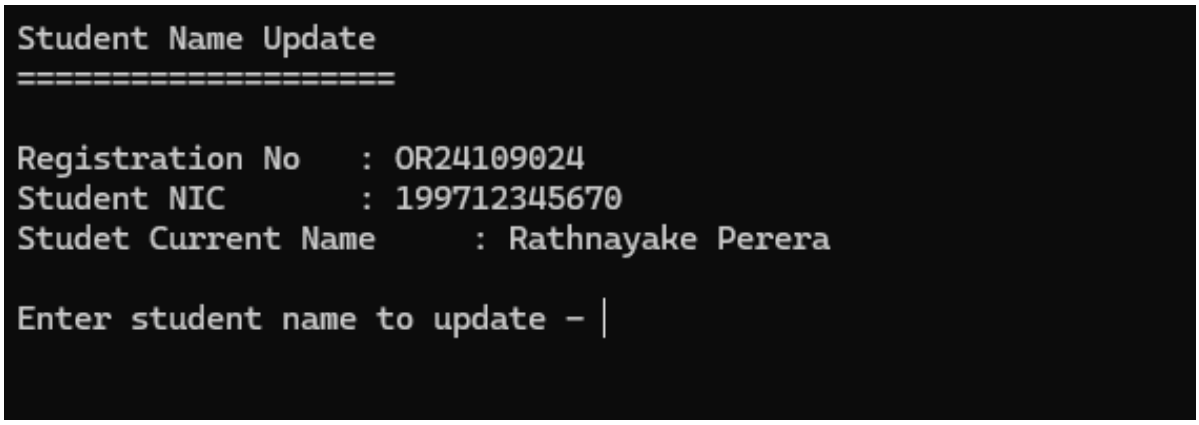


Figure 8 - Student Name Update

```
Student Name Update
===================

Registration No    : OR24109024
Student NIC        : 199712345670
Studet Current Name      : Rathnayake Perera

Enter student name to update - Aththanayake Perera


        Student name updated successfully...

Do you want to update another student details (Y/N):
```

Figure 9 - Student Name Update Successfully

1.2.2 Student NIC Update

If the user selects Student NIC (2), a window related to the Student NIC should be prompted. The system should check whether the entered NIC already exists in the system. If the NIC is found, the system should display a message indicating that the NIC already exists. If the NIC is not found, the system should allow the user to update the Student NIC.

```
Student NIC Update
==================

Registration No      : PR24110018
Student Name         : Perera Gunawardena
Student Current NIC  : 199601234566

Enter student NIC to update -
```

Figure 10 - Student NIC Update

Once the update has been completed successfully, the system should display a success message and prompt the user to choose whether to continue updating or return to the main menu.

```
Student NIC Update
==================

Registration No      : PR24110018
Student Name         : Perera Gunawardena
Student Current NIC  : 199601234566

Enter student NIC to update - 199601234567

        This student is already added to the system...

Do you want to update another student details (Y/N):
```

Figure 11 - Warning(Duplicate NIC)

1.3 View Student Profile

With this option, the user can view student details. First, the user must enter a valid Student ID. After the Student ID is entered, the system should check whether it exists in the database. If the Student ID has not been added yet, a message should be displayed as shown in Figure 14. If the Student ID is valid, the system should display the student details as shown in Figure 15,16 and 17.

The user can view 6 details about the student in this option: Registration No, Name, NIC, PRF Marks, DBMS Marks, and Student GPA. The Student GPA should be calculated based on the marks, using the GPA values mentioned in Figure 12. The system should calculate the GPA as:

(PRF Marks GPA + DBMS Marks GPA) / 2

Some students may be absent (indicated by -1) from the exam, and for some students, the exam may not have been conducted yet (indicated by -2).

Marks Range	GPA Value
90 – 100	4.25
80 – 89	4.00
75 – 79	3.70
70 – 74	3.30
65 – 69	3.00
60 – 64	2.70
55 – 59	2.30
50 – 54	2.00
45 – 49	1.70
40 – 44	1.30
30 – 39	1.00
20 – 29	0.70

Figure 12- GPA Value

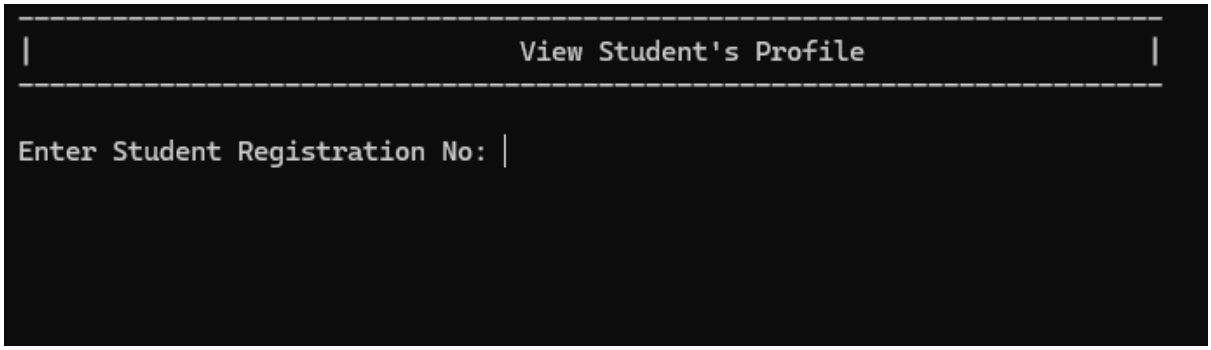


Figure 13 - View Student Profile

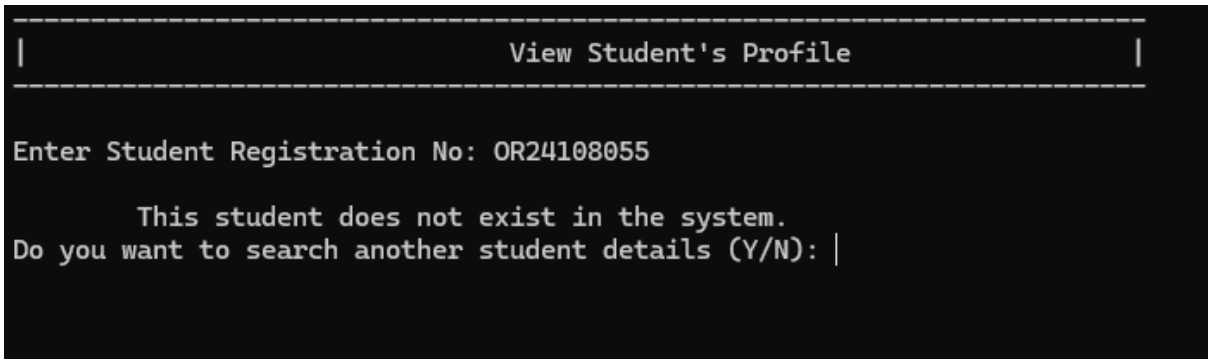


Figure 14 - Warning(Not Existing in the system)

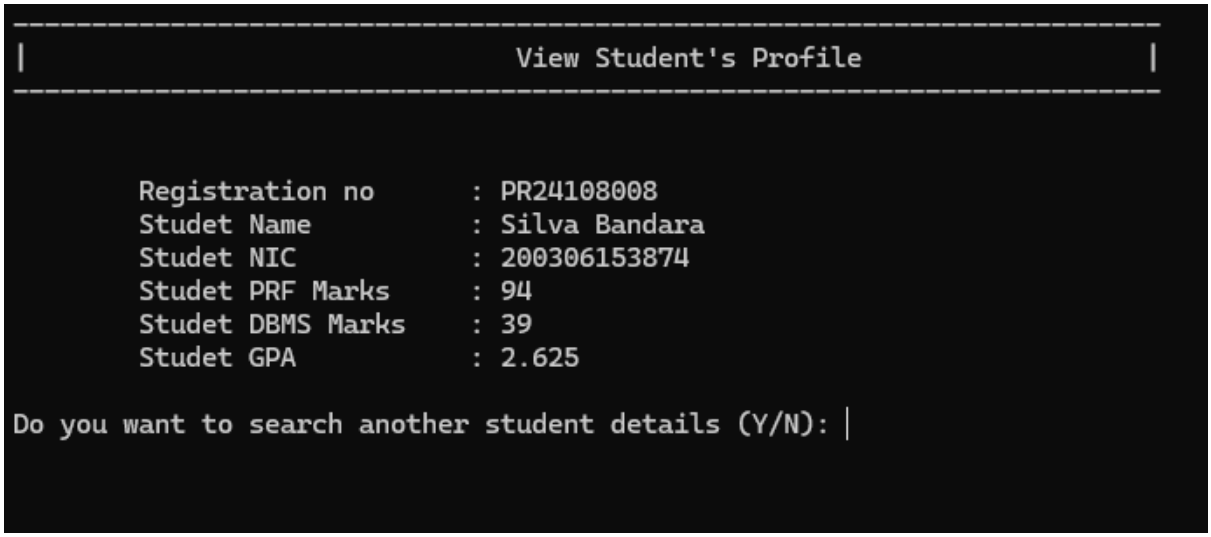


Figure 15 - Student Profile Data

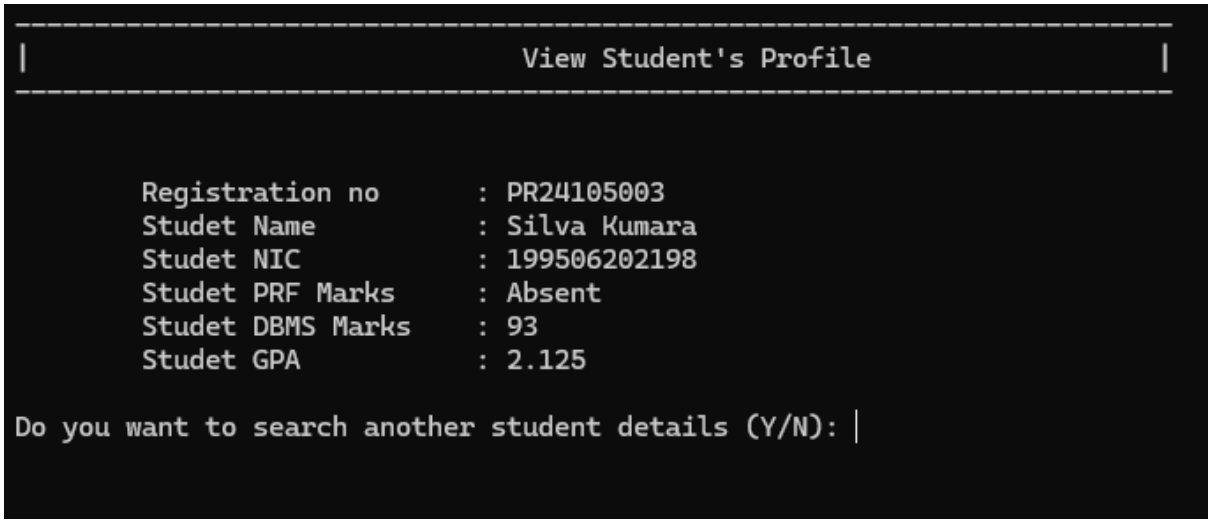


Figure 16 - Student Profile Data (Absent)

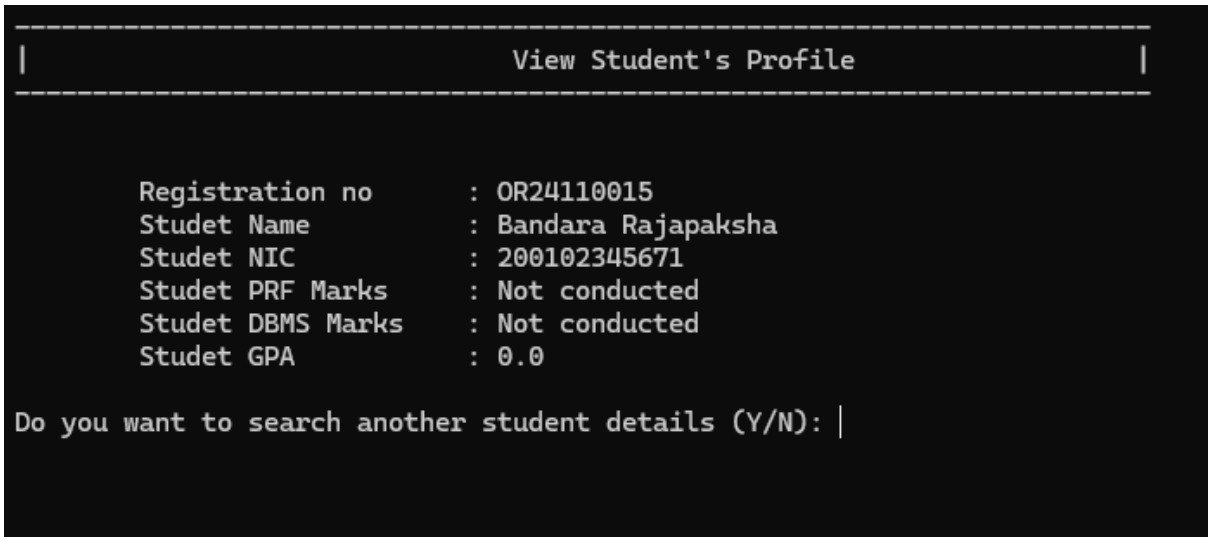


Figure 17 - Student Profile Data(Not Conducted Exam)

1.4 Delete Student Profile

With this option, a student can be deleted from the system. First, the user needs to enter the valid Student ID; otherwise, it should handle like previously. Once the user has entered a valid student id, the student can be deleted. Upon the successful deletion, the user should be prompted whether to delete another student or go back to the main menu.

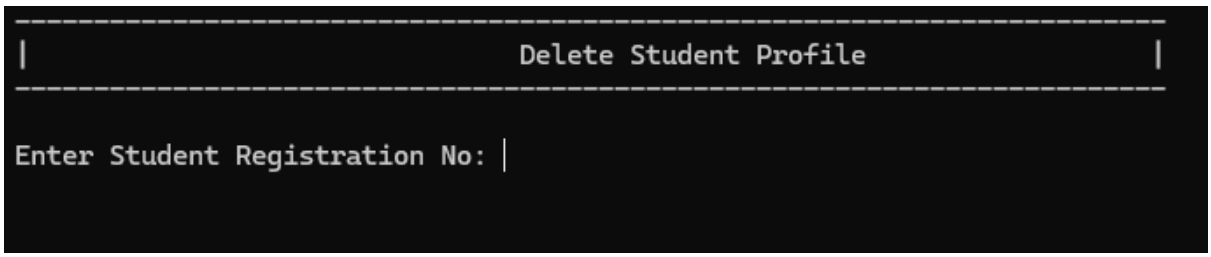


Figure 18 - Delete Student

```
------------------------------------------------------------------------
|                          Delete Student Profile                      |
------------------------------------------------------------------------

Enter Student Registration No: PR24109028

        This student does not exist in the system.

Do you want to delete another student profile (Y/N): |
```

Figure 19 - Warning(Not Existing in the system)

```
------------------------------------------------------------------------
|                          Delete Student Profile                      |
------------------------------------------------------------------------


        Registration no      : OR24108024
        Studet Name          : Gunawardena Ekanayake
        Studet NIC           : 200405143874
        Studet PRF Marks     : 71
        Studet DBMS Marks    : 88
        Studet GPA           : 3.65

Do you want to delete this student profile (Y/N): |
```

Figure 20- Display Student data before delete

```
------------------------------------------------------------------------
|                          Delete Student Profile                      |
------------------------------------------------------------------------


        Registration no      : OR24108024
        Studet Name          : Gunawardena Ekanayake
        Studet NIC           : 200405143874
        Studet PRF Marks     : 71
        Studet DBMS Marks    : 88
        Studet GPA           : 3.65

Do you want to delete this student profile (Y/N): Y

        Student was successfully deleted from the system.

Do you want to delete another student profile (Y/N): |
```

Figure 21 - Delete Successfully

[2] Batch Management

There are three options in Batch Management, and the user can select the corresponding option they wish to execute. Below is the Batch Management interface you should develop.



Figure 22 - Batch Management

2.1 Add Batch

The system stores three key details for each batch:

- Batch ID – Each batch has a unique number starting from 100.
- Batch Status – There are two possible statuses for a batch. The initial status of the batch is ENROLLMENTOPEN:
  - ENROLLMENTOPEN = 1
  - ENROLLMENTCLOSED = 0
- Student Count – The number of students enrolled in this batch.

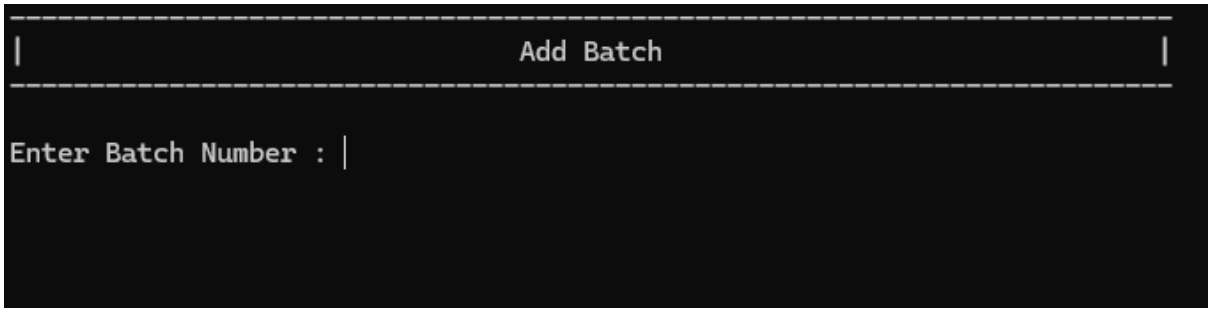


Figure 23 - Add Batch

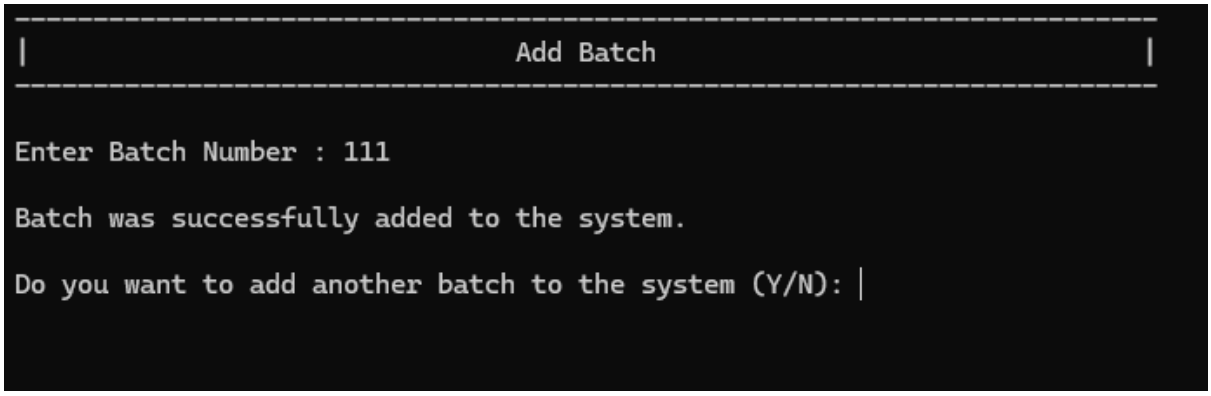


Figure 24 - Successfully add batch

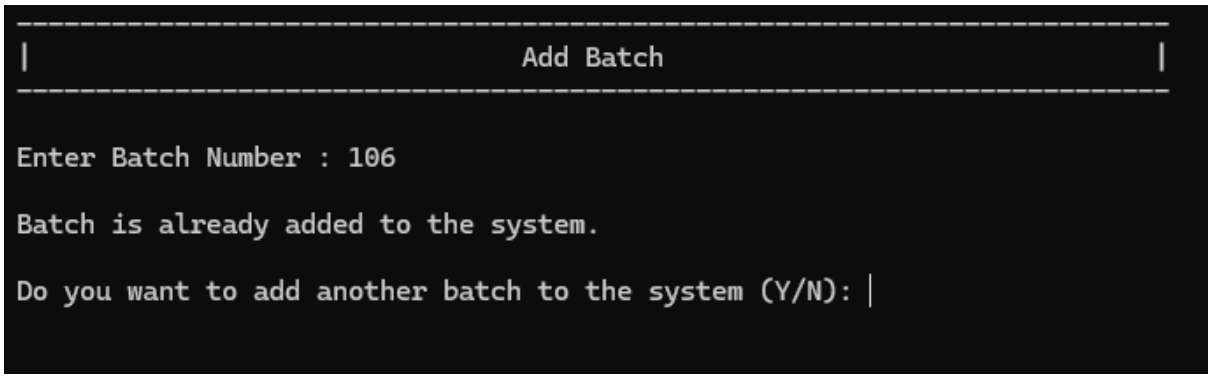


Figure 25 - Warning(Already added in the system)

2.2 Update Batch

With this option, the user can update previously added batch details (only the batch status). First, the user needs to find the batch by entering the batch ID. After the batch ID is entered, the system should check whether it exists in the system. If the batch ID has not been added yet, a message should be displayed: “This batch does not exist in the system.

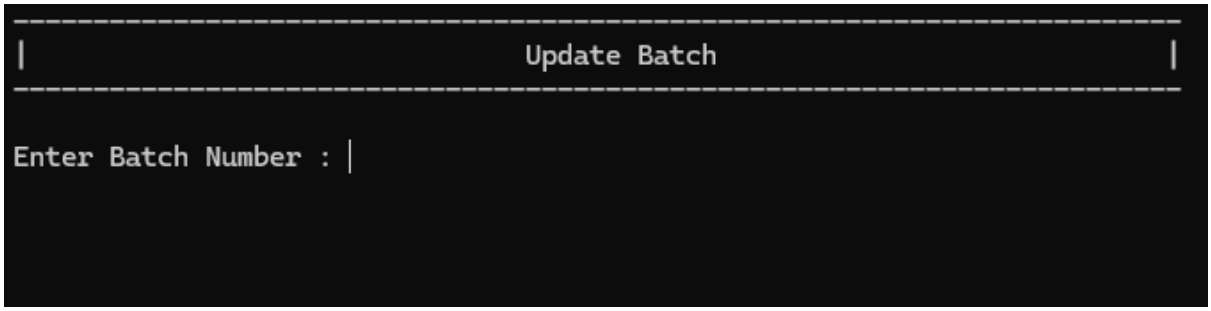


Figure 26- Update Batch

```
------------------------------------------------------------------------
|                              Update Batch                            |
------------------------------------------------------------------------

Enter Batch Number : 109
Current Status : ENROLLMENT OPEN

Do you want to change the batch status to ENROLLMENT CLOSED(Y/N) ? |
```

Figure 27 - Confirm batch status change

```
------------------------------------------------------------------------
|                              Update Batch                            |
------------------------------------------------------------------------

Enter Batch Number : 109
Current Status : ENROLLMENT OPEN

Do you want to change the batch status to ENROLLMENT CLOSED(Y/N) ? Y

        Batch Status updated successfully...

Do you want to update another batch details (Y/N): |
```

Figure 28 - Successfully change batch status

2.3 View Batches

This displays all the batches with details, including the batch name, student count, and status. The system should then prompt the user to choose whether they want to stay here or go back to the main menu.

```
------------------------------------------------------------------------
|                              View Batch                              |
------------------------------------------------------------------------


-------------------------------------------------
No    Batch No    Student Count  Status
-------------------------------------------------
1     105         25             ENROLLMENT CLOSED
-------------------------------------------------
2     106         25             ENROLLMENT CLOSED
-------------------------------------------------
3     107         25             ENROLLMENT CLOSED
-------------------------------------------------
4     108         25             ENROLLMENT CLOSED
-------------------------------------------------
5     109         25             ENROLLMENT OPEN
-------------------------------------------------
6     110         25             ENROLLMENT OPEN
-------------------------------------------------

Do you want to go to the home page (Y/N): |
```

Figure 29 - View Batch Data

[3] Grade Management

There are two options in Grade Management, and the user can select the corresponding option they wish to execute. Below is the Grade Management interface you should develop.

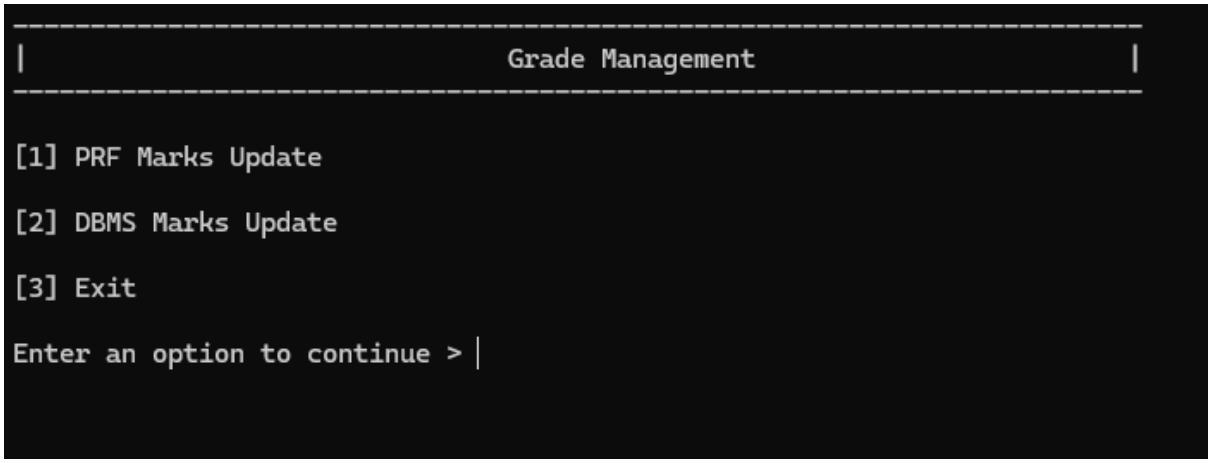


Figure 30 - Grade Management

3.1 PRF Marks Update

First, the user needs to find the student whose PRF marks should be updated via Student ID. (Invalid Student ID's should be handled like previously) Once the user has entered a valid student ID, it is going to display the current details. Then it prompts for new marks. Again, marks should be validated here. Once the update has been done successfully, it should prompt whether to continue updating or go back to the main menu.

The marks format is as follows: if the student is absent, the mark should be **-1**; otherwise, marks must be within the range of **0 to 100**.

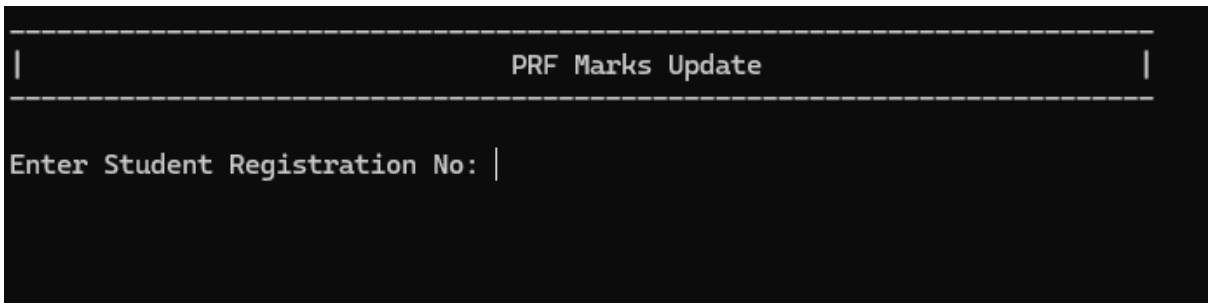


Figure 31 - PRF Marks Update

```
------------------------------------------------------------------------
|                            PRF Marks Update                          |
------------------------------------------------------------------------


        Registration no      : PR24105001
        Studet Name          : Gunawardena Weerasinghe
        Studet NIC           : 199501012345

This student has already completed the PRF module.
        PRF Marks : 85

Do you want to update this student's PRF marks?
```

Figure 32 - Update already added mark

```
------------------------------------------------------------------------
|                            PRF Marks Update                          |
------------------------------------------------------------------------


        Registration no      : PR24105001
        Studet Name          : Gunawardena Weerasinghe
        Studet NIC           : 199501012345

This student has already completed the PRF module.
        PRF Marks : 85

Do you want to update this student's PRF marks?Y
        Enter PRF Marks : 87

        This student PRF Marks updated successfully...

Do you want to update another student PRF marks (Y/N):
```

Figure 33 - Update already added mark successfully

```
------------------------------------------------------------------------
|                            PRF Marks Update                          |
------------------------------------------------------------------------


        Registration no      : OR24110020
        Studet Name          : Silva Rajapaksha
        Studet NIC           : 200412345678
        Enter PRF Marks :
```

Figure 34 - Update new marks

```
------------------------------------------------------------------------
|                            PRF Marks Update                          |
------------------------------------------------------------------------


        Registration no      : OR24110020
        Studet Name          : Silva Rajapaksha
        Studet NIC           : 200412345678
        Enter PRF Marks : 78

        This student PRF Marks updated successfully...

Do you want to update another student PRF marks (Y/N): |
```

Figure 35 - Update new marks successfully

```
------------------------------------------------------------------
|                          PRF Marks Update                      |
------------------------------------------------------------------


       Registration no      : PR24105003
       Studet Name          : Silva Kumara
       Studet NIC           : 199506202198

This student was absent from the exam. You can update the marks if they participate in it..
       Enter PRF Marks : |
```

Figure 36 - Absent mark update

```
------------------------------------------------------------------
|                          PRF Marks Update                      |
------------------------------------------------------------------


       Registration no      : PR24105003
       Studet Name          : Silva Kumara
       Studet NIC           : 199506202198

This student was absent from the exam. You can update the marks if they participate in it..
       Enter PRF Marks : 69

       This student PRF Marks updated successfully...

Do you want to update another student PRF marks (Y/N): |
```

Figure 37 - Absent mark update successfully

3.2 DBMS Marks Update

First, the user needs to find the student whose DBMS marks should be updated via Student ID. (Invalid Student ID's should be handled like previously) Once the user has entered a valid student ID, it is going to display the current details. Then it prompts for new marks. Again, marks should be validated here. Once the update has been done successfully, it should prompt whether to continue updating or go back to the main menu.

The marks format is as follows: if the student is absent, the mark should be **-1**; otherwise, marks must be within the range of **0 to 100**.

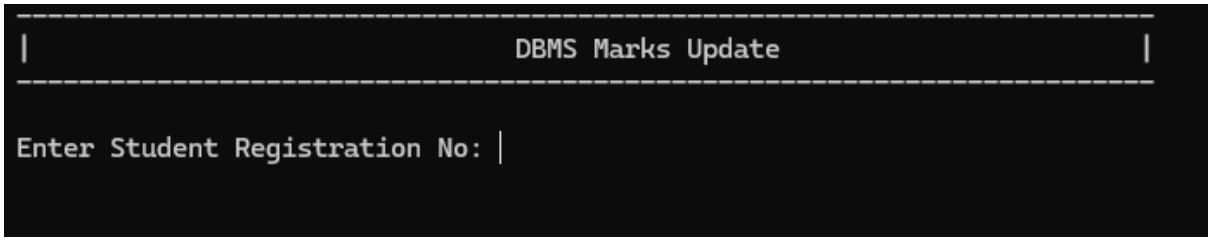


Figure 38 - - DBMS Marks Update

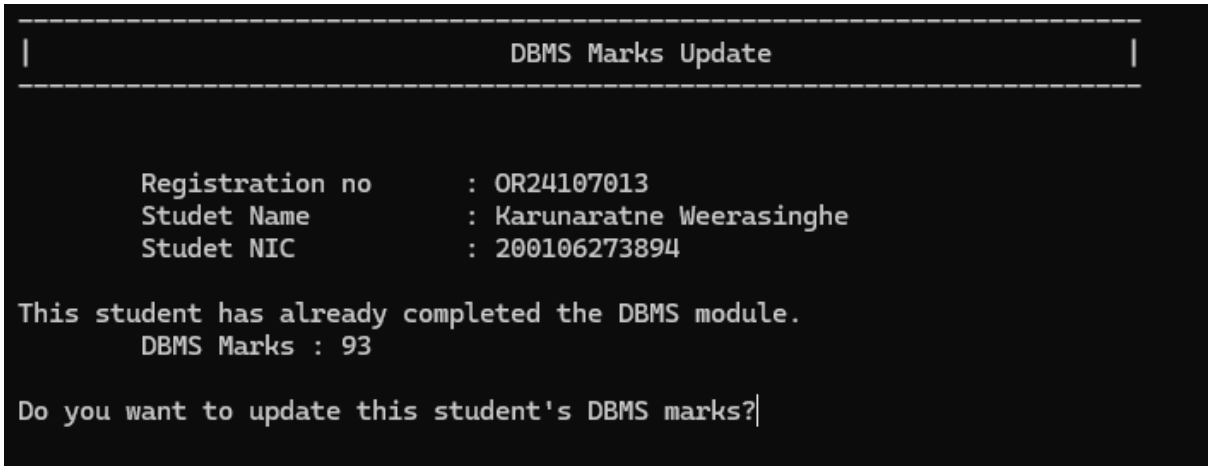


Figure 39 - Update already added mark

```
------------------------------------------------------------------------
|                            DBMS Marks Update                         |
------------------------------------------------------------------------


        Registration no      : OR24107013
        Studet Name          : Karunaratne Weerasinghe
        Studet NIC           : 200106273894

This student has already completed the DBMS module.
        DBMS Marks : 93

Do you want to update this student's DBMS marks?Y
        Enter DBMS Marks : 95

        This student DBMS Marks updated successfully...

Do you want to update another student DBMS marks (Y/N):
```

Figure 40 - Update already added mark successfully

```
------------------------------------------------------------------------
|                            DBMS Marks Update                         |
------------------------------------------------------------------------


        Registration no      : PR24110010
        Studet Name          : Rajapaksha Jayasinghe
        Studet NIC           : 200608789012
        Enter DBMS Marks :
```

Figure 41 - Update new marks

```
------------------------------------------------------------------------
|                            DBMS Marks Update                         |
------------------------------------------------------------------------


        Registration no      : PR24110010
        Studet Name          : Rajapaksha Jayasinghe
        Studet NIC           : 200608789012
        Enter DBMS Marks : 78

        This student DBMS Marks updated successfully...

Do you want to update another student DBMS marks (Y/N):
```

Figure 42 - Update new marks successfully

```
------------------------------------------------------------------------
|                            DBMS Marks Update                         |
------------------------------------------------------------------------


        Registration no      : PR24105005
        Studet Name          : Rathnayake Herath
        Studet NIC           : 199511258739

This student was absent from the exam. You can update the marks if they participate in it..
        Enter DBMS Marks :
```

Figure 43 - Absent mark update

```
------------------------------------------------------------------------
|                            DBMS Marks Update                         |
------------------------------------------------------------------------


        Registration no      : PR24105005
        Studet Name          : Rathnayake Herath
        Studet NIC           : 199511258739

This student was absent from the exam. You can update the marks if they participate in it..
        Enter DBMS Marks : 56

        This student DBMS Marks updated successfully...

Do you want to update another student DBMS marks (Y/N):
```

Figure 44 - Absent mark update successfully

[4] Report Management

There are three options in Report Management, and the user can select the corresponding option they wish to execute. Below is the Report Management interface you should develop.

```
------------------------------------------------------------------------
|                            Report Generator                          |
------------------------------------------------------------------------

[1] Student Registration Report

[2] Batch-wise Student Report

[3] Industry Training Eligibility Report

[4] Exit

Enter an option to continue >
```

4.1 Student Registration Report

This displays all students with their details in alphabetical order by student name, including the batch name, student registration number, PRF marks, DBMS marks, and student GPA. The system should then prompt the user to choose whether to stay on this page or return to the main menu.

```
-------------------------------------------------------------------------------------------------------------------------
                                        Student Registration Report
-------------------------------------------------------------------------------------------------------------------------


-------------------------------------------------------------------------------------------------------------------------
No   Registration No   Student Name                    NIC                 PRF Marks      DBMS Marks     GPA
-------------------------------------------------------------------------------------------------------------------------
1    OR24105023        Abeysekera Jayasinghe           199501012345              85              85          4.00
2    PR24107022        Abeysekera Perera               199503153872              39              39          1.00
3    PR24109008        Abeysekera Rajapaksha           199506202198              -1              -1          0.00
4    PR24110005        Abeysekera Rajapaksha           199509102983              72              72          3.30
5    PR24105017        Abeysekera Silva                199511258739              44              44          1.30
6    PR24106009        Abeysekera Silva                199512303498              91              91          4.25
7    OR24107011        Abeysekera Silva                199502183764              60              60          2.70
8    OR24108023        Abeysekera Silva                199504223198              38              38          1.00
9    OR24109015        Abeysekera Silva                199508153210              95              95          4.25
10   PR24108009        Abeysekera Weerasinghe          199510293417              49              49          1.70
11   OR24107012        Bandara Gunawardena             199601102375              -1              -1          0.00
12   PR24110022        Bandara Herath                  199604182938              67              67          3.00
13   PR24106016        Bandara Karunaratne             199606243879              23              23          0.70
14   OR24110015        Bandara Rajapaksha              199608142178              58              58          2.30
15   OR24105020        Bandara Rathnayake              199610312475              88              88          4.00
16   PR24105022        De Silva Dias                   199611173452              81              81          4.00
17   PR24106018        De Silva Silva                  199603293481              73              73          3.30
18   PR24106010        Dias Fernando                   199605083217              29              29          0.70
19   OR24108011        Dias Rajapaksha                 199607232198              62              62          2.70
20   PR24107005        Dias Senanayake                 199609192375              -1              -1          0.00
21   OR24106012        Ekanayake Bandara               199701212483              79              79          3.70
22   PR24109016        Ekanayake Bandara               199703132487              53              53          2.00
23   PR24110007        Ekanayake Bandara               199706253478              94              94          4.25
24   OR24108014        Ekanayake Jayasinghe            199708083298              47              47          1.70
25   PR24110021        Ekanayake Kumara                199710243651              35              35          1.00
26   OR24105015        Ekanayake Rathnayake            199712152983              93              93          4.25
27   PR24109009        Fernando Bandara                199702182734              15              15          0.00
28   OR24107015        Fernando Dias                   199704293187              -1              -1          0.00
29   OR24105013        Fernando Perera                 199705142375              82              82          4.00
30   PR24106017        Fernando Perera                 199709083751              45              45          1.70
31   PR24108001        Fernando Rajapaksha             199801032874              88              88          4.00
32   OR24108024        Gunawardena Ekanayake           199803232871              23              23          0.70
33   OR24110014        Gunawardena Ekanayake           199806193428              79              79          3.70
34   PR24107001        Gunawardena Kumara              199808013764              37              37          1.00
35   PR24109003        Gunawardena Perera              199810242374              -1              -1          0.00
36   PR24110008        Gunawardena Perera              199812302984              68              68          3.00
37   OR24106015        Gunawardena Rathnayake          199802152348             100             100          4.25
38   PR24105010        Gunawardena Silva               199805213471              59              59          2.30
39   OR24108015        Gunawardena Silva               199807172398              29              29          0.70
40   PR24105001        Gunawardena Weerasinghe         199811283472              92              92          4.25
41   PR24109018        Gunawardena Weerasinghe         199901122471              12              12          0.00
42   PR24107006        Herath Abeysekera               199903052984              77              77          3.70
43   PR24108004        Herath Abeysekera               199906213874              38              38          1.00
44   PR24110002        Herath Fernando                 199908093412              66              66          3.00
45   PR24105016        Herath Gunawardena              199910273894               9               9          0.00
46   PR24106007        Herath Jayasinghe               199912153482              84              84          4.00
47   PR24108022        Herath Karunaratne              199902202394              51              51          2.00
48   OR24110019        Herath Karunaratne              199904163874              32              32          1.00
49   OR24109014        Herath Kumara                   199907293481              -1              -1          0.00
50   OR24108012        Herath Perera                   199911083479              97              97          4.25
51   OR24107020        Herath Rajapaksha               200001112374              95              95          4.25
52   OR24106020        Herath Weerasinghe              200003143478              -1              -1          0.00
53   PR24109005        Herath Wijesinghe               200006293874              63              63          2.70
54   PR24106002        Jayasinghe Bandara              200008103471              88              88          4.00
55   OR24105019        Jayasinghe Dias                 200010252984              32              32          1.00
56   PR24106022        Jayasinghe Silva                200012043894              76              76          3.70
```

4.2 Batch-wise Student Report

First, the user should select the batch for which they want to view student data. After selection, the system will display all registered students in the chosen batch, including the batch name, student registration number, PRF marks, DBMS marks, and student GPA. The system will then prompt the user to choose whether to remain on this page or return to the main menu.

```
--------------------------------------------------------------------------------
|                          Batch-wise Student Report                           |
--------------------------------------------------------------------------------

[1] 105 Batch

[2] 106 Batch

[3] 107 Batch

[4] 108 Batch

[5] 109 Batch

[6] 110 Batch

[7] Exit

Enter an option to continue > 
```

```
--------------------------------------------------------------------------------------------------------------------
                                        105 Batch Student Report
--------------------------------------------------------------------------------------------------------------------

--------------------------------------------------------------------------------------------------------------------
No   Registration No   Student Name                        NIC                PRF Marks      DBMS Marks      GPA
--------------------------------------------------------------------------------------------------------------------
1    OR24105023        Abeysekera Jayasinghe               199501012345             85              85         4.00
2    PR24105017        Abeysekera Silva                    199511258739             44              44         1.30
3    OR24105020        Bandara Rathnayake                  199610312475             88              88         4.00
4    PR24105022        De Silva Dias                       199611173452             81              81         4.00
5    OR24105015        Ekanayake Rathnayake                199712152983             93              93         4.25
6    OR24105013        Fernando Perera                     199705142375             82              82         4.00
7    PR24105010        Gunawardena Silva                   199805213471             59              59         2.30
8    PR24105001        Gunawardena Weerasinghe             199811283472             92              92         4.25
9    PR24105016        Herath Gunawardena                  199910273894              9               9         0.00
10   OR24105019        Jayasinghe Dias                     200010252984             32              32         1.00
11   PR24105009        Karunaratne Jayasinghe              200007153871             23              23         0.70
12   OR24105014        Kumara Abeysekera                   200104103874             17              17         0.00
13   PR24105004        Kumara Herath                       200107202983             29              29         0.70
14   PR24105025        Kumara Karunaratne                  200210293874             86              86         4.00
15   PR24105007        Rajapaksha Herath                   200406293874             43              43         1.30
16   OR24105024        Rajapaksha Senanayake               200402203874             77              77         3.70
17   PR24105005        Rathnayake Herath                   200510083874             63              63         2.70
18   PR24105008        Senanayake Karunaratne              200512345678             68              68         3.00
19   PR24105002        Senanayake Silva                    199812346789             88              88         4.00
20   PR24105003        Silva Kumara                        199909876543             -2              -2         0.00
21   PR24105021        Silva Perera                        199812346789             -2              -2         0.00
22   OR24105012        Silva Rathnayake                    200608789012             -2              -2         0.00
23   OR24105011        Weerasinghe Rajapaksha              200412345678             -2              -2         0.00
24   PR24105018        Weerasinghe Silva                   199601234567             -2              -2         0.00
25   PR24105006        Wijesinghe Bandara                  200412345678             -2              -2         0.00

Do you want to another batch report (Y/N): 
```

4.3 Industry Training Eligibility Report

In this option, the system will display all students who are eligible for Industry Training based on the selected criteria. The displayed details will include the batch name, student registration number, PRF marks, DBMS marks, and student GPA. The system will then prompt the user to choose whether to remain on this page or return to the main menu.

Industry Training Eligibility Criteria:

1. GPA must be greater than 3.25
2. Both PRF and DBMS marks must be greater than 50

```
                              Industry Training Eligibility Student Report

No   Registration No   Student Name                    NIC                PRF Marks      DBMS Marks     GPA
1    PR24105001        Gunawardena Weerasinghe         199501012345          85             66          3.50
2    PR24105007        Rajapaksha Herath               199502183764          60             88          3.35
3    PR24105009        Karunaratne Jayasinghe          199508153210          95             79          3.98
4    OR24105015        Ekanayake Rathnayake            199610312475          88             92          4.13
5    PR24105017        Abeysekera Silva                199603293481          73             84          3.65
6    OR24105023        Abeysekera Jayasinghe           199706253478          94             98          4.25
7    PR24106008        Kumara Gunawardena              199806193428          79             80          3.85
8    PR24106025        Rathnayake Kumara               199911083479          97            100          4.25
9    PR24107004        Silva Ekanayake                 200008103471          88             74          3.65
10   PR24107006        Herath Abeysekera               200012043894          76             82          3.85
11   PR24107007        Rathnayake Fernando             200002193874          97             66          3.63
12   OR24107011        Abeysekera Silva                200101232984          90             80          4.13
13   OR24107013        Karunaratne Weerasinghe         200106273894          81             93          4.13
14   PR24107016        Weerasinghe Gunawardena         200112213874          67             85          3.50
15   PR24107021        Kumara Jayasinghe               200201013874          70             77          3.50
16   OR24107024        Kumara Weerasinghe              200208083471          60            100          3.48
17   PR24108001        Fernando Rajapaksha             200212183471          86             79          3.85
18   PR24108003        Perera Wijesinghe               200204123894          91             87          4.13
19   OR24108012        Herath Perera                   200302273874          80             83          4.00
20   OR24108014        Ekanayake Jayasinghe            200305123874          88             92          4.13
21   PR24108016        Rajapaksha Perera               200401153874         100             91          4.25
22   OR24108020        Kumara Herath                   200410213874          90             68          3.63
23   PR24108022        Herath Karunaratne              200402203874          77             66          3.35
24   OR24108024        Gunawardena Ekanayake           200405143874          71             88          3.65
25   PR24109001        Weerasinghe Kumara              200501023874          92             67          3.63
26   PR24109002        Rajapaksha Abeysekera           200503193471          68             91          3.63
27   PR24109004        Karunaratne Silva               200508213471          85             73          3.65
28   PR24109007        Silva Fernando                  200502123874          91             76          3.98
29   PR24109008        Abeysekera Rajapaksha           200504153471          70             88          3.65
30   OR24109011        Weerasinghe Jayasinghe          200203456782          72             79          3.50
31   OR24109012        Silva Karunaratne               200305678901          89             80          4.00
32   OR24109014        Herath Kumara                   199511223344          81             92          4.13
33   OR24109015        Abeysekera Silva                200412345678          77             68          3.35
34   PR24109016        Ekanayake Bandara               200512345678          68            100          3.63
35   PR24109018        Gunawardena Weerasinghe         199812346789          88             83          4.00
36   OR24109020        Silva Dias                      200608789012          90             62          3.48
37   OR24109023        Jayasinghe Silva                199902345678          92             78          3.98
38   OR24109024        Rathnayake Perera               199712345670          62             85          3.35
39   PR24109025        Silva Ekanayake                 200102345671         100             56          3.28

Do you want to go to homepage (Y/N):
```

Guidelines

- You may use any PRF knowledge to implement this coursework, such as flow control, loops, methods, and arrays.

- Except for arrays, you are not allowed to use any other Java data structure classes from the Java SE framework, such as ArrayList, LinkedList, etc., or any third-party libraries or frameworks.

- You are not allowed to create additional classes other than the one containing the main method. However, you may create as many methods as needed within that single class.

- Use the Scanner class to get input from the command-line interface.

- All validations that have been mentioned in this document should be implemented in your program.

- Please refer to the following Java code file, which includes the method structure and sample data initialized in arrays to help you get started.

  StudentManagementSystem.java

- The code to clear the command line from inside a Java application is as follows. You can use this code when you need to clear the command line.

```
public final static void clearConsole() {
        try {
                final String os = System.getProperty("os.name");
                if (os.contains("Windows")) {
                   new ProcessBuilder("cmd", "/c", "cls").inheritIO().start().waitFor();
                } else {
                   System.out.print("\033[H\033[2J");
                   System.out.flush();
                }
        } catch (final Exception e) {
                e.printStackTrace();
                // Handle any exceptions.
        }
}
```

- Following two lines of code are used to clear the last few lines of output in the console. The first line moves the cursor up lines, and the second line clears those lines and any subsequent lines that were printed after them. You can change the highlighted number for clear lines according to your need.

```
// Move the cursor up five lines
System.out.print("\033[5A");
// Clear the lines
System.out.print("\033[0J");
```

- This coursework is a **mandatory requirement** for your academics. Therefore, you **must complete and submit it before the deadline**.